Q1 - 24 January - Shift 2

Let $S = \{\theta \in [0, 2\pi) : \tan(\pi\cos\theta) + \tan(\pi\sin\theta) = 0\}$

Then $\sum_{\theta \in S} \sin^2\left(\theta + \frac{\pi}{4}\right)$ is equal to

Space for your notes:

Q2 - 29 January - Shift 2

The set of all values of λ for which the equation

$\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$

(1) $[-2, -1]$

(2) $\left[-2, -\frac{3}{2}\right]$

(3) $\left[-1, -\frac{1}{2}\right]$

(4) $\left[-\frac{3}{2}, -1\right]$

Space for your notes:

Q3 - 30 January - Shift 1

If $\tan 15° + \frac{1}{\tan 75°} + \frac{1}{\tan 105°} + \tan 195° = 2a$,

then the value of $\left(a + \frac{1}{a}\right)$ is :

(1) 4

(2) $4 - 2\sqrt{3}$

(3) 2

(4) $5 - \frac{3}{2}\sqrt{3}$

Space for your notes:

Q4 - 30 January - Shift 1

If the solution of the equation $\log_{\cos x} \cot x + 4\log_{\sin x} \tan x = 1, x \in \left(0, \frac{\pi}{2}\right)$, is $\sin^{-1}\left(\frac{\alpha+\sqrt{\beta}}{2}\right)$, where α, β are integers, then $\alpha+\beta$ is equal to:

(1) 3
(2) 5
(3) 6
(4) 4

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (2) **Q2** (4) **Q3** (1) **Q4** (4)

Q1 (2)

$\tan(\pi \cos \theta) + \tan(\pi \sin\theta) = 0$

$\tan(\pi \cos \theta) = -\tan(\pi \sin\theta)$

$\tan(\pi \cos \theta) = \tan(-\pi \sin\theta)$

$\pi \cos \theta = n\pi - \pi \sin\theta$

$\sin\theta + \cos\theta = n$ where $n \in I$

possible values are $n = 0, 1$ and -1 because

$-\sqrt{2} \le \sin\theta + \cos\theta \le \sqrt{2}$

Now it gives $\theta \in \left\{0, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{7\pi}{4}, \frac{3\pi}{2}, \pi\right\}$

So $\sum_{\theta \in S} \sin^2\left(\theta + \frac{\pi}{4}\right) = 2(0) + 4\left(\frac{1}{2}\right) = 2$

Q2 (4)

$\lambda = \cos^2 2x - 2\sin^4 x - 2\cos^2 x$

convert all in to cos x.

$\lambda = (2\cos^2 x - 1)^2 - 2(1 - \cos^2 x)^2 - 2\cos^2 x$

$= 4\cos^4 x - 4\cos^2 x + 1 - 2(1 - 2\cos^2 x + \cos^4 x) - 2\cos^2 x$

$= 2\cos^4 x - 2\cos^2 x + 1 - 2$

$= 2\cos^4 x - 2\cos^2 x - 1$

$= 2\left[\cos^4 x - \cos^2 x - \frac{1}{2}\right]$

$= 2\left[\left(\cos^2 x - \frac{1}{2}\right)^2 - \frac{3}{4}\right]$

$\lambda_{max} = 2\left[\frac{1}{4} - \frac{3}{4}\right] = 2 \times \left(-\frac{2}{4}\right) = -1$ (max Value)

$\lambda_{min} = 2\left[0 - \frac{3}{4}\right] = -\frac{3}{2}$ (Minimum Value)

So, Range $= \left[-\frac{3}{2}, -1\right]$

Q3 (1)

Option (1)

$\tan 15° = 2 - \sqrt{3}$

$\frac{1}{\tan 75°} = \cot 75° = 2 - \sqrt{3}$

$\frac{1}{\tan 105°} = \cot(105°) = -\cot 75° = \sqrt{3} - 2$

$\tan 195° = \tan 15° = 2 - \sqrt{3}$

$\therefore \quad 2(2 - \sqrt{3}) = 2a \Rightarrow a = 2 - \sqrt{3}$

$\Rightarrow \quad a + \frac{1}{a} = 4$

Q4 (4)

$\log_{\cos x} \cot x + 4\log_{\sin x} \tan x = 1$

$\Rightarrow \dfrac{\ln\cos x - \ln\sin x}{\ln\cos x} + 4\dfrac{\ln\sin x - \ln\cos x}{\ln\sin x} = 1$

$\Rightarrow (\ln\sin x)^2 - 4(\ln\sin x)(\ln\cos x) + 4(\ln\cos x)^2 =$

$\Rightarrow \ln\sin x = 2\ln\cos x$

$\Rightarrow \sin^2 x + \sin x - 1 = 0 \Rightarrow \sin x = \dfrac{-1+\sqrt{5}}{2}$

$\therefore \alpha + \beta = 4$

Correct option (4)